

## SAFETY DATA SHEET

Creation Date 22-September-2009

Revision Date 24-December-2021

Revision Number 5

### 1. Identification

**Product Name** 1,4-Diethylbenzene

**Cat No. :** AC114150000; AC114150050; AC114150250

**CAS-No** 105-05-5  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                          |            |
|------------------------------------------|------------|
| <b>Flammable liquids</b>                 | Category 3 |
| <b>Skin Corrosion/Irritation</b>         | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b> | Category 2 |
| <b>Aspiration Toxicity</b>               | Category 1 |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
Flammable liquid and vapor  
May be fatal if swallowed and enters airways  
Causes skin irritation

Causes serious eye irritation



### Precautionary Statements

#### Prevention

Keep container tightly closed

Ground/bond container and receiving equipment

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Use only non-sparking tools

Take precautionary measures against static discharges

Wash face, hands and any exposed skin thoroughly after handling

Wear protective gloves/protective clothing/eye protection/face protection

#### Response

IF SWALLOWED: Immediately call a POISON CENTER/doctor

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Do NOT induce vomiting

If skin irritation occurs: Get medical advice/attention

If eye irritation persists: Get medical advice/attention

Wash contaminated clothing before reuse

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

#### Storage

Store locked up

Store in a well-ventilated place. Keep cool

#### Disposal

Dispose of contents/container to an approved waste disposal plant

### Other Hazards

Toxic to aquatic organisms

Toxic to aquatic life with long lasting effects

## 3. Composition/Information on Ingredients

| Component        | CAS-No   | Weight % |
|------------------|----------|----------|
| p-Diethylbenzene | 105-05-5 | 98       |

## 4. First-aid measures

### General Advice

If symptoms persist, call a physician.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).

### Ingestion

Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call

a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

**Most important symptoms/effects** None reasonably foreseeable. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Notes to Physician** Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

**Unsuitable Extinguishing Media** No information available

**Flash Point** 56 °C / 132.8 °F

**Method -** CC (closed cup)

**Autoignition Temperature** 430 °C / 806 °F

### Explosion Limits

**Upper** .00%

**Lower** .80%

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Combustible material. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions** Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Storage.** Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Strong oxidizing agents.

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### **Respiratory Protection**

No protective equipment is needed under normal use conditions.

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                         |                              |
|-----------------------------------------|------------------------------|
| <b>Physical State</b>                   | Liquid                       |
| <b>Appearance</b>                       | Colorless                    |
| <b>Odor</b>                             | Strong                       |
| <b>Odor Threshold</b>                   | No information available     |
| <b>pH</b>                               | No information available     |
| <b>Melting Point/Range</b>              | -43 °C / -45.4 °F            |
| <b>Boiling Point/Range</b>              | 184 °C / 363.2 °F @ 760 mmHg |
| <b>Flash Point</b>                      | 56 °C / 132.8 °F             |
| <b>Method -</b>                         | CC (closed cup)              |
| <b>Evaporation Rate</b>                 | No information available     |
| <b>Flammability (solid,gas)</b>         | Not applicable               |
| <b>Flammability or explosive limits</b> |                              |
| <b>Upper</b>                            | .00%                         |
| <b>Lower</b>                            | .80%                         |
| <b>Vapor Pressure</b>                   | 140.52 Pa @ 25 °C            |

|                                        |                          |
|----------------------------------------|--------------------------|
| Vapor Density                          | 4.64                     |
| Specific Gravity                       | 0.862                    |
| Solubility                             | slightly soluble         |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | 430 °C / 806 °F          |
| Decomposition Temperature              | No information available |
| Viscosity                              | 3.6 mPas @ 20°C          |
| Molecular Formula                      | C10 H14                  |
| Molecular Weight                       | 134.22                   |
| Refractive index                       | 1.495                    |

## 10. Stability and reactivity

|                                  |                                                                                                                          |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                               |
| Stability                        | Stable under normal conditions.                                                                                          |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. Excess heat. Exposure to light. Incompatible products. |
| Incompatible Materials           | Strong oxidizing agents                                                                                                  |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                                                  |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                 |
| Hazardous Reactions              | None under normal processing.                                                                                            |

## 11. Toxicological information

### Acute Toxicity

**Product Information** Product does not present an acute toxicity hazard based on known or supplied information

#### Component Information

| Component        | LD50 Oral                      | LD50 Dermal | LC50 Inhalation                            |
|------------------|--------------------------------|-------------|--------------------------------------------|
| p-Diethylbenzene | OECD 401 (Rat)<br>> 2000 mg/kg | Not listed  | OECD 403 (Rat)<br>> 5000 mg/m <sup>3</sup> |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes and skin                                                              |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component        | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------|----------|------------|------------|------------|------------|------------|
| p-Diethylbenzene | 105-05-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** Category 1

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component        | Freshwater Algae                                            | Freshwater Fish                         | Microtox        | Water Flea                             |
|------------------|-------------------------------------------------------------|-----------------------------------------|-----------------|----------------------------------------|
| p-Diethylbenzene | Pseudokirchnerella<br>Subcapitata<br>EC50 = 1.9 mg/L, 72 hr | Oryzias Latipes<br>LC50 = 1.8 mg/L, 96h | EC50 = 250 mg/L | Daphnia Magna<br>LC50 = 32 mg/L, 24 hr |

**Persistence and Degradability** May persist based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component        | log Pow      |
|------------------|--------------|
| p-Diethylbenzene | 4.06 @ 25 °C |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN2049  
 Proper Shipping Name DIETHYLBENZENE  
 Hazard Class 3  
 Packing Group III

### TDG

UN-No UN2049  
 Proper Shipping Name DIETHYLBENZENE  
 Hazard Class 3  
 Packing Group III

### IATA

UN-No UN2049  
 Proper Shipping Name DIETHYLBENZENE  
 Hazard Class 3  
 Packing Group III

### IMDG/IMO

UN-No UN2049  
 Proper Shipping Name DIETHYLBENZENE  
 Hazard Class 3  
 Packing Group III

## 15. Regulatory information

### International Inventories

| Component | CAS-No | DSL | NDL | TSCA | TSCA Inventory notification - | EINECS | ELINCS | NLP |
|-----------|--------|-----|-----|------|-------------------------------|--------|--------|-----|
|-----------|--------|-----|-----|------|-------------------------------|--------|--------|-----|

|                  |          |   |   |   | Active-Inactive |           |   |   |
|------------------|----------|---|---|---|-----------------|-----------|---|---|
| p-Diethylbenzene | 105-05-5 | - | X | X | ACTIVE          | 203-265-2 | - | - |

| Component        | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------|----------|-------|----------|------|------|------|------|-------|-------|
| p-Diethylbenzene | 105-05-5 | X     | KE-10437 | X    | X    | X    | -    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH****Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component        | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| p-Diethylbenzene | 105-05-5 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component        | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| p-Diethylbenzene | 105-05-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Revision Summary**

SDS sections updated. 2. 3. 4. 6. 7. 8. 9. 10. 11. 12. 16.

**Disclaimer**

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**End of SDS**